

TUGAS 2

STRUKTUR SISTEM OPERASI



Dosen pengampu :
Triawan Adi Cahyanto, M.K

Disusun Oleh:
Muhammad Syarifuddin (2210651057)

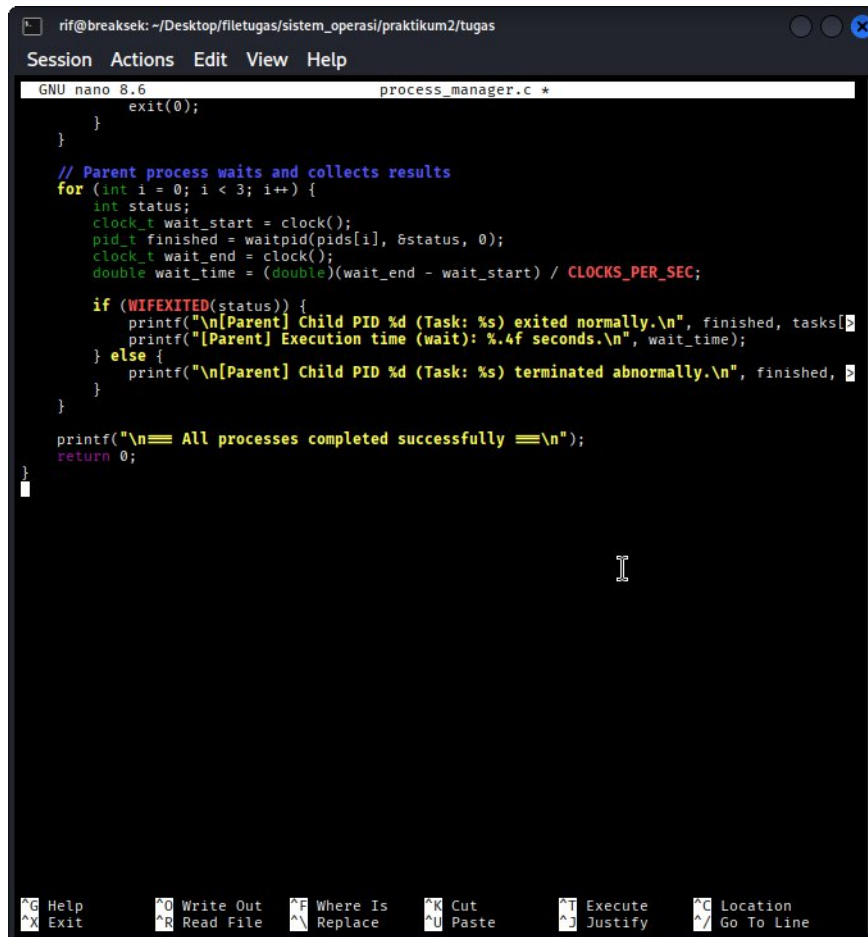
PRODI TEKNIK INFORMATIKA
FAKULTAS TEKNIK
UNIVERSITAS MUHAMMADIYAH JEMBER

DAFTAR ISI

TUGAS 2	1
DAFTAR ISI.....	2
1. Sistem Manajemen Proses	2
2. Chat Application dengan IPC.....	4

1. Sistem Manajemen Proses

1.1. Membuat file process_manager.c



```
GNU nano 8.6 process_manager.c *
    exit(0);
}

// Parent process waits and collects results
for (int i = 0; i < 3; i++) {
    int status;
    clock_t wait_start = clock();
    pid_t finished = waitpid(pids[i], &status, 0);
    clock_t wait_end = clock();
    double wait_time = (double)(wait_end - wait_start) / CLOCKS_PER_SEC;

    if (WIFEXITED(status)) {
        printf("\n[Parent] Child PID %d (Task: %s) exited normally.\n", finished, tasks[i]);
        printf("[Parent] Execution time (wait): %.4f seconds.\n", wait_time);
    } else {
        printf("\n[Parent] Child PID %d (Task: %s) terminated abnormally.\n", finished, tasks[i]);
    }
}

printf("\n== All processes completed successfully ==\n");
return 0;
}
```

1.2. Mencompile dan menjalankan program

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi/praktikum2/tugas
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$ sudo nano process_manager.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$ gcc -o process_manager process_manager.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$ ./process_manager
== Simple Process Management Simulation ==

[Child 1] PID: 4553 | Task: Sorting
Sorted array: 1 2 5 7 8 9
[Child 1] Finished in 0.0001 seconds.

[Child 2] PID: 4554 | Task: Searching
Value 5 found at index 2
[Child 2] Finished in 0.0001 seconds.

[Parent] Child PID 4553 (Task: Sorting) exited normally.
[Parent] Execution time (wait): 0.0000 seconds.

[Child 3] PID: 4555 | Task: Calculation
Average value: 8.00
[Child 3] Finished in 0.0001 seconds.

[Parent] Child PID 4554 (Task: Searching) exited normally.
[Parent] Execution time (wait): 0.0000 seconds.

[Parent] Child PID 4555 (Task: Calculation) exited normally.
[Parent] Execution time (wait): 0.0000 seconds.

== All processes completed successfully ==

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$
```

2. Chat Application dengan IPC

2.1. Membuat file chat_shm.c

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi/praktikum2/tugas
Session Actions Edit View Help
GNU nano 8.6 chat_shm.c *
#define _XOPEN_SOURCE 700
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/mman.h>
#include <semaphore.h>
#include <time.h>
#include <signal.h>
#include <errno.h>
#include <sys/stat.h>
#include <sys/types.h>

#define SHM_NAME "/chat_shm_example_v1"
#define SEM_NAME "/chat_sem_example_v1"

#define MAX_MESSAGES 100
#define MESSAGE_LEN 256
#define USERNAME_LEN 32

typedef struct {
    pid_t pid;
    char username[USERNAME_LEN];
    time_t ts;
    char text[MESSAGE_LEN];
} chat_msg_t;

typedef struct {
    int initialized; // 1 jika sudah diinisialisasi
    int user_count; // berapa proses yang sedang terhubung
    int start; // index pesan pertama (oldest)
    int count; // jumlah pesan saat ini
    chat_msg_t messages[MAX_MESSAGES];
} chat_shm_t;

static chat_shm_t *shm_ptr = NULL;
static sem_t *mutex = NULL;
static int shm_fd = -1;
static char username[USERNAME_LEN];

void cleanup_and_exit(int exit_code);

void sigint_handler(int sig) {
    (void)sig;
    cleanup_and_exit(EXIT_SUCCESS);
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^I Execute  ^C Location
^X Exit      ^R Read File ^A Replace   ^U Paste     ^D Justify  ^_ Go To Line
```

2.2. Mencompile dan menjalankan program

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi/praktikum2/tugas
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$ gcc chat_shm.c -o chat_shm -pthread -lrt

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$ ./chat_shm rafi
Tersambung ke chat. Username: rafi | PID: 4723 | Users online: 1

Menu:
1) Kirim pesan
2) Lihat history
3) Exit
Pilih (1-3): 1
Ketik pesan: Hai aku rafi
Pesan terkirim.

Menu:
1) Kirim pesan
2) Lihat history
3) Exit
Pilih (1-3): 2

— Chat History (last 1 messages) —
[2025-10-21 19:10:49] rafi (PID 4723): Hai aku rafi
— end —

Menu:
1) Kirim pesan
2) Lihat history
3) Exit
Pilih (1-3): 3
Keluar ...
Shared memory dihapus.
Semaphore dihapus.

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi/praktikum2/tugas]
$
```